

## A new Source-Pathway-Receptor based model for identification and evaluation of potentially contaminated sites

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A longstanding lag in environmental oversight has led to the accumulation of a vast number of potentially contaminated sites (PCS) worldwide. As investigation of all locations is unrealistic, effective approaches are required to manage these sites and minimize associated risks.

Multicriteria decision analysis (MCDA)-based risk assessment is widely applied to the management of contaminated sites; however, only a limited number of studies specifically address PCS. Moreover, existing approaches often rely on heterogeneous sets of attributes (criteria) that do not consistently reflect the underlying principles of groundwater flow and contaminant transport. This work addresses this gap by identifying more practical and physically meaningful attributes derived from parameters of simple analytical transport models.

The proposed methodology, termed  $SPR_M$ , is based in the Source-Pathway-Receptor (SPR) paradigm and considers direct contact and groundwater as the primary exposure routes. A Choquet integral-based aggregation function is employed to capture interactions among pathway and receptor attributes. To evaluate performance,  $SPR_M$  was applied to 20 contaminated sites, using groundwater plume dimensions (length, area, and width) as primary risk indicators. Statistical evaluation, including Spearman rank correlation coefficients and complementary tests, demonstrated significant strong to very strong positive correlations between plume dimensions and  $SPR_M$  scores. A comparative analysis involving three additional similar models (from literature) applied to the same dataset further highlighted the importance of selecting attributes informed by transport-process understanding. The  $SPR_M$  framework is both simple and conceptually coherent, and its structure allows for straightforward recalculation of results when conceptual site models are updated.

Overall, the methodology provides a practical decision-support tool for prioritizing investigation efforts at PCS that pose the highest potential risks to receptors. Ongoing work includes expanding the dataset to encompass a broader group of contaminants and integrating attributes that account for climate-change-related impacts on contaminant fate and transport.

Keywords: Potentially contaminated sites, Source Pathway Receptor Model, Multicriteria decision analysis, Environmental risk assessment